
Multi-Agent Appeal Games in Aggregate Disambiguation Systems

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Abstract

When a panel reaches a provisional decision, multiple agents may decide whether to trigger another evaluation. We study these appeal games in aggregate disambiguation systems and ask when private incentives produce the socially preferred level of escalation. We measure the welfare loss from missed appeals and unnecessary re-evaluations using the gap between the action selected by the agents and the action preferred by a social planner. We show that funding the next panel is not sufficient to align incentives: no fixed bond and reward can work across all workloads when social decision loss and private transaction value vary independently. Under a calibrated single-crossing model, we derive a reward that implements the planner’s escalation threshold. With several potential appellants, equilibrium can produce both missed appeals and redundant submissions, while increasing bonds can exclude capital-constrained agents. We also separate corrective credibility into the probability that an erroneous decision is appealed and the probability that the successor panel corrects it. The results identify the conditions and measurements required for multi-agent appeal mechanisms to approximate socially optimal escalation.

Keywords: optimal stopping; mechanism design; appeals; aggregate disambiguation; focal coordination; optimistic democracy

1 Introduction

A finite-panel decision system has two distinct economic problems. First, someone must decide whether the present evidence is sufficient or whether another evaluation should be purchased. Second, once a panel has been selected, its members must find it privately rational to perform the intended evaluation and to coordinate on the resulting decision. These problems are related, but they are not interchangeable. A mechanism may fund every additional panel and still request one at the wrong histories; it may also stop at the right time while rewarding validators for a cheaper correlated shortcut.

Aggregate Disambiguation Systems provide a statistical target for the panel decision, and adaptive economic aggregate disambiguation provides a planner benchmark for choosing among stop, re-evaluate, and regenerate actions. This paper studies the binary stop–re-evaluate specialization. At each public history, the planner continues exactly when the expected reduction in decision loss exceeds the resource and latency cost of another panel. A decentralized protocol instead delegates that option to private appellants whose beliefs, transaction exposure, capital, and race costs need not equal the social primitives.

The central question is therefore an implementation question: under what conditions does rational appeal behavior reproduce the planner’s stopping rule? We answer it in four steps.

First, we specialize the classical performance-difference lemma to finite-horizon panel stopping. The regret of an equilibrium appeal policy is the expected sum of Bellman action gaps at the histories it occupies. This prices premature stopping and excessive escalation in the same welfare units and makes clear that appeal volume is not an implementation metric.

Second, we separate panel funding from incentive alignment. No fixed bond and reward can implement every workload when social loss and private transaction value vary independently. We sharpen this separation result with an exact joint-dispersion formula: implementation regret depends on the joint probability that social and private thresholds disagree, weighted by the planner’s loss from the disagreement.

Third, we give a positive conditional result. If a calibrated scalar success belief is sufficient for both the planner and an exclusive appellant, and the planner’s action is single-crossing in that belief, an explicit reward implements the planner threshold. This result reveals what a payout ratio actually calibrates. A successful return of one and a half times the posted principal, for example, creates a private success threshold of two thirds for a risk-neutral, unconstrained appellant with no additional cost; it does not establish that the social threshold is two thirds.

Fourth, we add the strategic frictions suppressed by the exclusive-appellant model. A first-valid-transaction entry game has regimes with no entry, full entry, and interior mixing. In its interior, missed appeals and redundant submissions coexist. Priority competition can dissipate the entire exclusive rent, while a growing bond can preserve the nominal payoff threshold yet exclude otherwise willing appellants. Inside the panel, we separately factor corrective credibility into the probability of an appeal following an error and the probability that the successor panel corrects it. Under an explicit belief model, that measurable product determines how strongly independent execution acts as a focal signal.

Optimistic Democracy is the application case. Its fee formulas fund additional work and impose meaningful costs on contention. Our results show which further objects—workload-specific loss, successor-panel value, appellant beliefs, ordering costs, capital availability, and corrective credibility—must be measured before calling the deployed stopping rule economically optimal. The conclusions are conditional on the declared model and revision-scoped implementation artifacts; they are not a semantic-truth claim or an empirical claim that current actors satisfy the assumed equilibrium.

The rest of the main paper presents the methodology, the analytic and numerical results, the protocol mapping, and the resulting design implications. Complete statements and proofs are collected in the appendices. The deterministic numerical design and reproduction instructions appear in the supplementary material.

2 Methodology

2.1 Planner benchmark and implementation regret

Fix a declared workload and a finite sequence of possible panel rounds. At public history h in round r , stopping incurs expected decision loss $Q_r^*(h, \text{Stop})$; continuing incurs the next panel’s resource and latency cost plus its optimally managed future loss, denoted $Q_r^*(h, \text{Continue})$. The planner’s continuation advantage is

$$G_r(h) = Q_r^*(h, \text{Stop}) - Q_r^*(h, \text{Continue}).$$

The planner continues when this quantity is positive and stops when it is negative. These values are workload-specific: changing the loss attached to an incorrect outcome can reverse the action without changing the panel or fee schedule.

An appeal mechanism and an equilibrium selection induce an aggregate stopping policy. For an action a , define its Bellman gap by

$$g_r(h, a) = Q_r^*(h, a) - \min_{a'} Q_r^*(h, a').$$

The finite-horizon performance-difference identity gives

$$\text{Reg}(\pi) = \mathbb{E}_\pi \left[\sum_{r=0}^{\tau} g_r(H_r, A_r) \right],$$

where τ is the stopping time induced by policy π . Thus only reached histories with the wrong action contribute regret, and their contribution is the social cost of that action mismatch. Appendix A states and proves the identity with the required information and integrability conditions.

2.2 Private appeal incentives

For a risk-neutral potential appellant i , let $q_i(h)$ be its subjective probability that an appeal is classified as successful, $B(h)$ the principal lost on failure, $R(h)$ the net reward on success, $m_i(h)$ its monitoring, execution, capital, and

submission cost, and $x_i(h)$ its private value from continuation. Its incremental expected payoff is

$$U_i(h) = q_i(h)R(h) - [1 - q_i(h)]B(h) + x_i(h) - m_i(h).$$

The protocol-defined success event, the event that another panel lowers population-decision loss, and the appellant's private value are distinct primitives. Exact payment for triggered work constrains an accounting flow; it does not force the sign of $U_i(h)$ to equal the sign of $G_r(h)$.

To quantify the resulting mismatch, consider one occupied public cell in which another panel reduces error probability by $\Delta > 0$ and costs $c > 0$. If transaction loss is L , the planner continues above $L^* = c/\Delta$. Holding the transfer terms fixed, suppose the appellant continues above a private-value threshold x^* . The cell's exact regret is

$$\Delta \mathbb{E}[|L - L^*| \mathbf{1}\{(L - L^*)(x - x^*) < 0\}].$$

This expression requires the joint law of social loss and private value; their separate marginals do not identify implementation quality. Appendix B gives the universal impossibility result and the full joint-dispersion proposition.

2.3 A calibrated single-crossing implementation

A one-dimensional payout can implement the planner only after imposing one-dimensional structure. Let z denote observable variables on which a transfer may depend and let q be a calibrated belief about the protocol's success event. Assume that, within each z , the planner continues exactly when q exceeds a social threshold q_z^* . Assume also one exclusive, risk-neutral trigger with known non-bond cost m_z , no additional private transaction term, bond $B_z > 0$, and net success reward R_z .

Setting

$$R_z = \frac{(1 - q_z^*)B_z + m_z}{q_z^*}$$

makes private payoff equal to $(B_z + R_z)(q - q_z^*)$. Private and social actions then agree at every value of q . The result is constructive but conditional: the calibrated signal must be sufficient for social continuation value, the threshold must be estimable, and the transfer must satisfy the protocol's separate budget, observability, and anti-collusion constraints. Appendix C contains the theorem and a closed-form quadratic regret formula for threshold misspecification.

2.4 Several appellants and capital

Permissionless appeals create a public-good and preemption problem. We study a symmetric reduced game with N potential appellants. If at least one submits, all receive continuation value v ; one uniformly selected first valid transaction additionally receives exclusive expected prize s ; every submitter pays race cost g . This isolates ordering and duplicated submission costs while remaining analytically soluble.

The equilibrium submission probability is zero when $g \geq v + s$, one when $g \leq s/N$, and otherwise is the unique interior solution of

$$v(1 - p)^{N-1} + s \frac{1 - (1 - p)^N}{Np} = g.$$

In the interior, both the probability of no appeal and the expected number of redundant submissions are strictly positive. A symmetric all-pay priority contest provides a complementary stress case in which expected priority expenditure equals the entire exclusive prize.

Bond scale adds a participation constraint. If an actor has liquid capital K_i , non-bond cost $m_i^0 + \lambda_i B$, private value x_i , and net success reward ζB , it must have $K_i \geq B$ and

$$q_i > \frac{1 + \lambda_i + (m_i^0 - x_i)/B}{1 + \zeta}.$$

For independent capital laws, the probability that at least one of N otherwise-willing actors can post the bond is

$$A_N(B) = 1 - \prod_{i=1}^N \mathbb{P}(K_i < B).$$

Appendix D provides the equilibrium proofs and separates real race cost from transfers.

2.5 Corrective credibility and focal coordination

The appeal decision is not the within-panel voting game. Validators can coordinate informatively by executing the task or uninformatively by following a common shortcut. Let

$$a = \mathbb{P}(\text{effective appeal} \mid \text{current decision is wrong})$$

and

$$c = \mathbb{P}(\text{corrective clawback} \mid \text{effective appeal and error}).$$

Their product $\chi = ac$ is corrective credibility: the conditional probability that an erroneous current majority is eventually corrected through appeal. This factorization distinguishes failures of monitoring and participation from failures of successor-panel correction.

Under a stated Laplacian belief over how many peers received a positive execution signal, the average payoff advantage of following that signal is proportional to χ and to the signal's mean direction. This is a conditional focality result, not an equilibrium-selection theorem: uninformative coordination equilibria can remain. The complete payoff model, the decomposition, and the limitations of the belief assumption are in Appendix E.

3 Results

The analysis yields a sharp separation between accounting adequacy and economic implementation. The main conclusions are summarized in Table 1. Each positive statement carries an explicit structural assumption; removing that assumption produces either a non-identification result or a measurable source of regret.

3.1 Threshold alignment and entry regimes

Figure 1 displays two exact analytic slices. The left panel compares the planner's threshold with the fixed private threshold induced by a successful return of one and a half times principal. Actions agree only when both thresholds place the history on the same side; the diagonal is exact implementation, and the horizontal line shows the bond-only private threshold of two thirds. The colored disagreement regions are the histories that receive positive weight in the Bellman-gap regret identity. Their welfare importance depends on occupancy and on the size of the planner gap, not merely on geometric area.

The right panel solves the preemption game for ten identical actors with continuation value one and exclusive prize 0.3. Very high submission cost produces no entry; sufficiently low cost produces full entry and nine redundant transactions; the interior region combines a positive probability of no appeal with positive expected duplication. Increasing the number of potential appellants does not mechanically remove this coexistence because it changes both free-riding and the race for the exclusive prize.

Threshold misspecification has a direct welfare interpretation. If the planner advantage is locally linear in calibrated success belief and that belief is uniform within a public cell, moving the implemented threshold by d produces one-cell regret proportional to $d^2/2$. More generally, regret integrates the absolute Bellman gap over the occupied belief mass lying between the private and social thresholds. The correct empirical target is therefore the distribution near the threshold, rather than a global average success rate.

The joint-dispersion result reaches the same conclusion without assuming that social action is indexed by success belief. When social loss and private transaction value cross different thresholds, the exact implementation penalty is the planner gap on their sign-disagreement event. A large private stake can cause excessive appeals for socially minor transactions; absence of private exposure can cause premature stopping for socially important ones. No workload-independent payout ratio distinguishes these cases.

3.2 Capital participation along an appeal ladder

Figure 2 separates the nominal bond-only threshold from capital feasibility. The left panel gives the validator-appeal bond in validator-work units for the nominal jury sizes. It rises from seven to 769 units, roughly a 110-fold increase. Under risk neutrality and zero extra costs, the private success threshold remains two thirds at every size because both downside and upside scale with the bond. The set of actors able to post that principal need not remain fixed.

Table 1. Economic object, result, and interpretation.

Object	Formal result	Design implication
Fixed bond and reward	No universal implementation when social loss and private value vary independently.	Funding another panel is not evidence that it is requested at the socially correct histories.
Calibrated exclusive trigger	An explicit success reward implements a single-crossing planner threshold.	The bond can fund work; the reward ratio selects a private threshold.
First-valid entry	The interior equilibrium has both missed appeals and redundant submissions.	Permissionless participation does not imply reliable, waste-free provision.
Growing bond	Capital-only availability is nonincreasing in the required principal.	A size-invariant nominal threshold can conceal a changing appellant population.
Corrective credibility	Correction probability factors into appeal incidence and conditional correction.	Participation and successor quality must be measured separately.

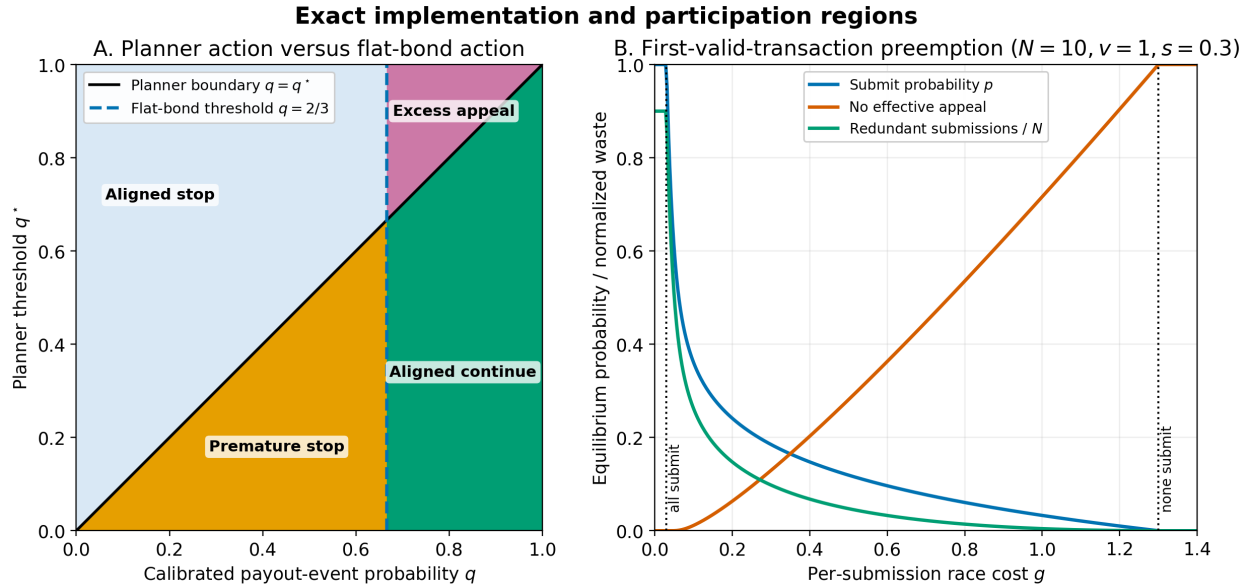


Figure 1. Two analytic implementation slices. Left: agreement and disagreement between a workload-specific planner threshold and the private threshold induced by the fixed payout ratio. Right: symmetric equilibrium of the first-valid-transaction entry game for the displayed parameter slice. These are deterministic illustrations, not fitted behavioral estimates.

The right panel illustrates this selection effect under an explicitly hypothetical lognormal capital distribution. Availability improves with the number of independent potential appellants but falls with the bond in every case. The curve is a sensitivity illustration, not an estimate of present validator capital or of live appeal probability. Beliefs, monitoring, sequencing, correlated capital, credit, and private value can reduce or increase realized participation relative to this capital-only layer.

3.3 Focal execution is conditional on correction

The within-panel model identifies a useful observable interface. A threat of reversal affects validator incentives only if an error is both appealed and corrected. Measuring appeal frequency alone conflates those channels. The identity $\chi = ac$ supports two separate diagnostics: low a points toward monitoring, private-value, capital, or entry frictions; low c points toward successor-panel quality, correlation, or the definition of successful appeal.

Under the auxiliary Laplacian belief, the mean action advantage from using the independent execution signal is $W\chi(2\mu - 1)$, where W is the round fee scale and μ is the mean direction of that signal. Corrective credibility therefore amplifies focal selection when the signal is informative on average. The result does not prove truthful execution, unique equilibrium, or robustness to collusion. It states exactly which empirical probability makes appeal-backed correction economically relevant to the round game.

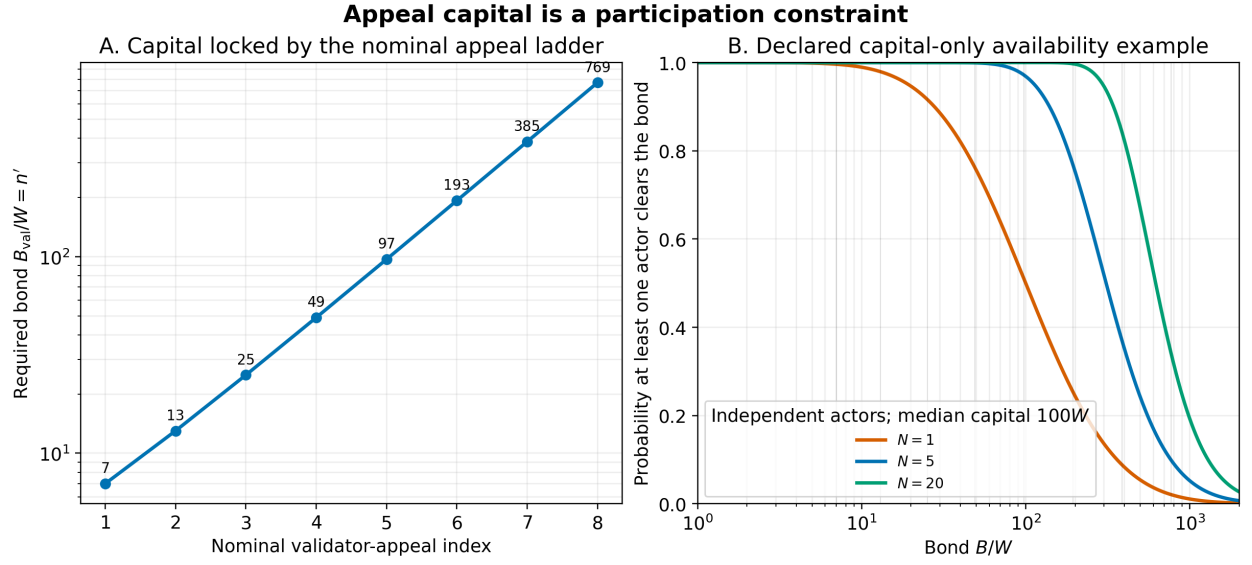


Figure 2. Bond scale and an illustrative capital constraint. Left: nominal validator-appeal bond along the panel ladder in units of validator work. Right: probability that at least one actor can post the bond under the stated hypothetical independent capital law. The distribution is used only for sensitivity analysis.

All displayed figures are generated from closed-form expressions. Their parameter choices and reproduction commands are documented in Supplementary Sections S1–S2; no random data or fitted protocol behavior enters these plots.

4 Applied Case: Optimistic Democracy

Optimistic Democracy uses provisional execution, validator panels, and bonded appeals. The application is useful because it exposes all three layers of the model: a panel produces a decision, an appellant decides whether to purchase a successor round, and the successor can overturn economic consequences for the previous majority. The mapping below is revision-scoped to the canonical state-machine, fee-simulator, and archived contract artifacts identified in Appendix F.

4.1 What the implemented bonds establish

For a validator appeal that purchases a successor jury of size n' , the modeled bond is

$$B_{\text{val}} = n'W,$$

where W is the per-validator fee unit. Leader and leader-timeout appeals include the relevant execution and validator-work terms,

$$B_{\text{lead}} = (\rho + 1)(L + mW), \quad B_{\text{lt}} = (\rho + 1)(L + n_{\text{cur}}W),$$

with leader fee L , multiplier ρ , and the relevant panel sizes. On the covered paths, these formulas fund triggered work and make repeated contention costly. Together with redistribution rules, they support accounting and bounded-griefing statements proved in Appendix F.

Those properties are valuable but narrower than optimal stopping. They do not identify the transaction-specific loss from finalizing an incorrect decision, the reduction in that loss produced by the successor round, or the appellant's private value and cost. Consequently, they do not by themselves determine the sign of the planner continuation advantage or show that a rational actor appeals whenever that sign is positive.

The implemented successful return of $1.5B$ means that principal is returned with net reward $B/2$. Ignoring additional costs and private transaction value, a risk-neutral actor appeals when its success belief exceeds $2/3$. With private cost m , the threshold rises to

$$\frac{2}{3} + \frac{2m}{3B}.$$

The economic question is not whether this algebra is correct; it is whether the social continuation threshold is also two thirds in each occupied workload cell. The calibration theorem shows how to choose a different reward when an aligned social threshold can be estimated. The impossibility result shows why a single universal threshold cannot be justified when transaction loss and private exposure vary independently.

4.2 Preemption, capital, and protocol scope

Only the first valid state-changing appeal purchases a given successor. Potential appellants can therefore face both a public-good problem and an ordering race. The first-valid game predicts missed appeals when submission costs are high relative to continuation and trigger value, full congestion when they are low, and coexistence of both failures in between. A live calibration must use the actual sequencing, revert, fee-bid, and information rules rather than the uniform-ordering reduction used for the analytic theorem.

Bond growth creates a second selection layer. The nominal payout threshold can remain constant while the liquid-capital requirement expands with the successor panel. Credit and delegation may mitigate that restriction; correlated balance sheets, concentration, and opportunity cost may strengthen it. Counting eligible addresses is therefore not a substitute for estimating effective capital availability at each tier.

The protocol mapping also has boundaries. The formal stopping result treats a fixed resolution evaluated by successor panels. Regeneration, multi-category outcomes, timeouts, replacement of inactive validators, stake-weighted sampling, repeated interaction, and state paths that change the resolution require enriched transition and game models. The pinned artifacts establish the formulas and state transitions used here, not invariance across future revisions or equivalence between every deployed path and the binary abstraction.

4.3 A deployment-oriented calibration protocol

The theory suggests a concrete empirical workflow.

1. **Declare workload cells and social loss.** Predeclare the transaction or application classes for which a common loss model is defensible. Estimate decision loss and latency cost per cell rather than importing one universal value.
2. **Estimate successor value.** For each reachable public history, estimate how the actual next-round experiment changes final decision risk and cost. This produces the planner action and Bellman gap. Statistical panel certificates inform this step but do not determine transaction loss.
3. **Estimate private behavior separately.** Measure appellant success beliefs, monitoring and submission costs, capital availability, private transaction exposure, and the effective number and information structure of competitors. The joint law of social loss and private value is required for the joint-dispersion regret.
4. **Measure the two correction channels.** On adjudicated errors, report both the probability of an effective appeal and the probability of correction conditional on that appeal. Their product is the credibility input to the focal-coordination model.
5. **Compare policies by Bellman-gap regret.** Report premature-stop mass, excessive-escalation mass, race waste, capital exclusion, statistical-risk feasibility, and the occupancy-weighted social loss of the induced action mismatches. Raw appeal count is not enough.
6. **Validate out of sample.** Commit any workload-dependent transfer schedule before testing it on held-out histories. Account for the selection problem that appeal outcomes are observed only where the previous mechanism made an appeal privately worthwhile.

This workflow answers a narrower question than whether validators execute the intended task. The stopping analysis determines when rational appellants purchase another experiment. Validator execution additionally requires an incentive-compatible information-acquisition model: the cost and detectability of true execution must be compared with imitation, shared shortcuts, and coordination on an uninformative focal point. Corrective credibility is one input to that model, not a complete proof of effort.

5 Discussion

The answer to whether a decentralized appeal mechanism “works” is conditional rather than binary. The mechanism implements economically optimal stopping when the induced equilibrium action matches the workload-specific Bellman action at occupied histories. A calibrated single-crossing environment with an exclusive trigger gives one sufficient construction. A fixed payout ratio, permissionless racing, and exact panel funding do not establish the required alignment without the additional workload, belief, ordering, and capital measurements.

This distinction clarifies the role of the three layers in the broader framework. Population-level panel analysis estimates whether another evaluation is likely to reproduce the declared aggregate decision. Adaptive economic analysis converts that statistical improvement into a social stop-or-continue benchmark using transaction-specific loss and real cost. The present mechanism analysis asks whether rational private behavior implements that benchmark. None of the layers can replace the inputs of the next: high panel clarity does not assign social loss, and a fully funded successor does not guarantee that somebody will call it at the right time.

The negative result is consequently constructive. It says which information a universal bond lacks. The joint-dispersion formula identifies the missing dependence between social loss and private value; the threshold theorem identifies the structural alignment needed for a scalar reward; the preemption game identifies the entry and congestion effects hidden by a representative appellant; and the capital formula identifies the participation margin hidden by a constant reward ratio. These objects can be estimated or stress-tested in a simulator before any live mechanism change.

Several extensions are especially important. First, a generative mechanism needs a joint stopping-and-resolution objective that prevents economic incentives from favoring an easy but low-fidelity replacement. Second, the reduced ordering game should be replaced with the actual sequencer, fee-bid, revert, timeout, and private-information rules. Third, capital constraints call for models of credit, delegation, concentration, and risk aversion. Fourth, multi-category decisions, inactive-validator replacement, weighted sampling, and stake feedback require a dynamic population model. Fifth, an effort-acquisition game should test when executing the task dominates imitation or correlated shortcuts. Finally, repeated interaction, side payments, strategic workload selection, and endogenous transaction creation can change both the planner loss distribution and the private-value distribution.

The present results supply a falsifiable benchmark for that program. A proposed appeal rule can be evaluated by estimating its equilibrium occupancy, computing the Bellman gaps it selects, and decomposing failures into premature stopping, excessive escalation, race waste, capital exclusion, and unsuccessful correction. For Optimistic Democracy, the current fee rules demonstrably finance covered successor work and impose contention costs. Whether they implement the socially optimal stopping rule is an empirical calibration question, not a consequence of the accounting identities alone.

Institutional Disclosure

This work was conducted within GenLayerLabs Research. GenLayerLabs develops GenLayer, whose Optimistic Democracy mechanism is analyzed as an application case. The general decision-theoretic and game-theoretic results do not depend on that implementation.

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Appendices: Proofs and Technical Results

A The Planner Benchmark and Its Decentralized Implementation

A.1 Bellman actions imported from Part 2

Fix a declared workload law Λ and Part 2's fixed-column, stop-re-evaluate specialization of the Bayes social objective. After round r , the observable history is $H_r = h$. Stopping incurs conditional population-decision loss

$$s_r(h) = \mathbb{E}[L(P)\mathbf{1}\{d_r \neq D_\infty(P)\} \mid H_r = h].$$

Continuing purchases the next experiment, with real resource and latency cost $a_{r+1}(h, H_{r+1})$, and then behaves optimally. Define the planner's two action values

$$\begin{aligned} Q_r^*(h, \text{Stop}) &= s_r(h), \\ Q_r^*(h, \text{Continue}) &= \mathbb{E}[a_{r+1}(h, H_{r+1}) + V_{r+1}(H_{r+1}) \mid H_r = h, \text{Continue}], \end{aligned}$$

where $V_r(h) = \min_a Q_r^*(h, a)$ and stopping is mandatory at the finite terminal index J . The social continuation advantage is

$$G_r(h) = Q_r^*(h, \text{Stop}) - Q_r^*(h, \text{Continue}).$$

Thus the planner re-evaluates when $G_r(h) > 0$, stops when $G_r(h) < 0$, and may randomize at equality. This is exactly the two-action specialization of Part 2's Bellman rule, with costs already sunk at h excluded.

A.2 Aggregate appeal policies

A decentralized appeal game, together with an equilibrium selection, induces an aggregate history-measurable policy π^{eq} : its continuation probability is the probability that the protocol receives an effective appeal at h . Comparing it with the planner is meaningful only when both policies belong to the same information and admissibility class. Private watcher signals must either be included in both information sets or in neither.

Definition A.1 (Implementation and implementation regret). An equilibrium appeal policy *implements* the planner when it places probability one on a Bellman-minimizing action at every history it reaches. Its welfare regret is

$$\text{Reg}_\Lambda(\pi^{\text{eq}}) = \mathcal{J}_\Lambda(\pi^{\text{eq}}) - \mathcal{J}_\Lambda(\pi^*) \geq 0.$$

For a hard statistical-risk objective, feasibility is reported separately: a cheap equilibrium that violates the risk constraint is not an implementation of the constrained planner.

The definition gives a scalar comparison, but the following identity locates the loss history by history. It is the finite-horizon, undiscounted stopping specialization of the performance-difference lemma of Kakade and Langford [7]; the adaptation below uses Part 2's cost convention and a random terminal time.

Lemma A.2 (Performance difference for AE-ADS stopping). *For a reached history h and action a , define the nonnegative Bellman gap*

$$g_r(h, a) = Q_r^*(h, a) - V_r(h).$$

Let A_r be the action induced by any admissible policy π at a reached round and let τ be its stopping time. Under the integrability and measurability conditions of Part 2,

$$\boxed{\mathcal{J}_\Lambda(\pi) - \mathcal{J}_\Lambda(\pi^*) = \mathbb{E}_{\Lambda, \pi} \left[\sum_{r=0}^{\tau} g_r(H_r, A_r) \right].}$$

Costs common to every policy, including the initial panel already purchased, cancel from both sides.

Proof. For a continuation action, the definition of $Q_r^\star$ gives

$$g_r(H_r, \text{Continue}) = \mathbb{E}[a_{r+1} + V_{r+1}(H_{r+1}) \mid H_r, \text{Continue}] - V_r(H_r).$$

For the stopping action at τ , $g_\tau(H_\tau, \text{Stop}) = s_\tau(H_\tau) - V_\tau(H_\tau)$. Sum these expressions along the path generated by π and apply the tower property. Consecutive $V_r(H_r)$ terms telescope. The remaining terms are the expected continuation costs plus the conditional terminal loss under π , minus $\mathbb{E}[V_0(H_0)]$, which is the optimal conditional objective under $\pi^\star$. Averaging over the workload and initial history proves the identity. $\square$

Corollary A.3 (Exact implementation and the two failure modes). *Regret is zero if and only if the induced action has zero Bellman gap almost surely at every policy-occupied history. If every reached comparison is strict, then*

$$\text{Reg}_\Lambda(\pi) = \mathbb{E}_{\Lambda, \pi} \left[\sum_{r=0}^{\min\{\tau, J-1\}} |G_r(H_r)| \mathbf{1}\{A_r \neq A_r^\star\} \right].$$

A disagreement with the planner is either premature stopping ($G_r > 0$ but nobody appeals) or excessive escalation ($G_r < 0$ but an appeal occurs).

This identity is the economic interface between Parts 2 and 4. A model of rational actors is useful to the series precisely to the extent that it identifies the occupancy probabilities and action mismatches on its right-hand side.

B Private Appeals and a Workload-Universal Impossibility

B.1 The appellant's private payoff

At history h , let S_h be the event that the protocol classifies an appeal as successful. For potential appellant i , define:

- $q_i(h) = \mathbb{P}_i(S_h \mid h, \text{appeal})$, its subjective success probability (calibration is an additional property);
- $B(h) > 0$, the posted principal lost on failure;
- $R(h) \geq 0$, the *net* success reward, excluding return of the principal;
- $m_i(h) \geq 0$, monitoring, execution, capital, and submission cost;
- $x_i(h)$, expected private transaction utility from inducing the continuation, relative to not appealing.

For a risk-neutral appellant, the expected incremental payoff is

$$U_i(h) = q_i(h)R(h) - (1 - q_i(h))B(h) + x_i(h) - m_i(h). \quad (1)$$

The success event in this equation is protocol-defined. In GenLayer it is a comparison of full outcome categories; it is not the latent event $d_{r+1} = D_\infty$.

With one exclusive trigger, continuation occurs when its $U_i(h)$ is positive. A frictionless reduced form with several actors might instead use $\max_i U_i(h) > 0$. Section D shows why an actual multi-agent equilibrium need not obey that shortcut.

B.2 Funding is not implementation

Proposition B.1 (No workload-universal fixed appeal rule). *Consider a reachable history where continuation has positive real cost and strictly reduces terminal population-inconsistency probability. Over a model class in which social decision loss and private transaction utility may vary independently, no fixed pair (B, R) makes the sign of (1) coincide with the sign of $G_r(h)$ for every admissible workload and private-utility profile, even when B exactly funds the work triggered by the appeal.*

Proof. Hold the public history, transition law, appellant beliefs, costs, and transfer rule fixed. Because continuation has positive cost and strictly reduces error, scaling the social loss $L(P)$ across its Bellman break-even value reverses the planner's action without changing any private payoff. Conversely, holding the social problem fixed while moving $x_i(h)$ across the private break-even value reverses the appellant's action without changing $G_r(h)$. The signs must therefore differ on some admissible profile. Exact work funding constrains an accounting flow; it does not identify either social loss or private value. $\square$

Remark B.2 (Three objects that must not be collapsed). The probability that an appeal is classified as successful, the probability that another panel reduces population-decision loss, and the private value of changing a transaction are different primitives. They can be related by a calibrated model, but none is a definition of either of the others.

The proposition is a separation fact, not an argument against appeals. Its quantitative content appears once a joint law for social loss and private value is declared.

B.3 Joint social-loss and private-value dispersion

Fix one occupied public cell z . Suppose one more panel reduces the probability of a population-inconsistent terminal decision by $\Delta_z > 0$, has optimized incremental social cost $c_z > 0$, and changes no other social term. For decision loss L , the planner advantage is

$$G_z(L) = \Delta_z L - c_z = \Delta_z (L - L_z^*), \quad L_z^* = \frac{c_z}{\Delta_z}.$$

Hold the appellant's payout belief and transfer terms fixed and write its payoff as $U_z(x) = u_z + x$, where x is private transaction value. Its private threshold is $x_z^* = -u_z$.

Proposition B.3 (Joint-dispersion implementation regret). *Let (L, x) have any declared joint law with $\mathbb{E}|L - L_z^*| < \infty$ and zero probability of either threshold tie. If the planner continues exactly when $L > L_z^*$ and the exclusive appellant appeals exactly when $x > x_z^*$, the conditional one-cell regret is*

$$\mathcal{R}_z = \Delta_z \mathbb{E}[|L - L_z^*| \mathbf{1}\{(L - L_z^*)(x - x_z^*) < 0\} \mid z]. \quad (2)$$

It is bounded by $0 \leq \mathcal{R}_z \leq \Delta_z \mathbb{E}[|L - L_z^| \mid z]$. If $0 < A \leq L_z^*$, L is uniform on $[L_z^* - A, L_z^* + A]$, and L is independent of the appellant's threshold action, then $\mathcal{R}_z = \Delta_z A/4$, regardless of the appellant's unconditional appeal probability.*

Proof. The two actions differ exactly when their centered indices have opposite signs. At such a history the Bellman action gap is $|G_z(L)| = \Delta_z |L - L_z^*|$; otherwise it is zero. This proves (2) by Lemma A.2, and dropping the indicator gives the bound. Under the final assumptions, each half of the uniform loss interval has probability 1/2 and conditional mean distance $A/2$. Independence weights the two mismatch halves by complementary appeal probabilities, giving $\Delta_z (A/2)(1/2) = \Delta_z A/4$. $\square$

Equation (2) is estimable from a declared joint workload/utility model. It also exposes why a one-dimensional payout probability can be insufficient: correlation between social loss and private value, not either marginal alone, determines the mismatch mass.

C A Positive Calibrated Implementation Result

C.1 Aligned single-crossing environments

Let z collect the public variables on which a transfer schedule may depend: round index, protocol status, eligible set, and preregistered workload class. Suppose an exclusive trigger additionally observes a calibrated scalar $q \in [0, 1]$ for the event on which the protocol pays the success reward. The information-matched planner is allowed the same signal. The restriction that makes one-dimensional implementation possible is the following.

Definition C.1 (Aligned single-crossing appeal environment). For every public cell z , there is a threshold $q_z^* \in (0, 1)$ such that the planner's continuation advantage satisfies

$$\text{sign } G(z, q) = \text{sign}(q - q_z^*).$$

The exclusive trigger is risk neutral, has no additional private transaction term, incurs known cost $m_z \geq 0$, and is offered a failure bond $B_z > 0$ and net success reward $R_z \geq 0$ that depend on z but not on its private q .

Single crossing is substantive. It can fail when the same protocol-success probability corresponds to different successor accuracy, latency, loss, or future options. In that case no one-dimensional bond threshold can encode the Bellman state.

Theorem C.2 (Calibrated threshold implementation). *In Definition C.1, fix any $B_z > 0$ and set*

$$R_z = \frac{(1 - q_z^*)B_z + m_z}{q_z^*}. \quad (3)$$

Then the trigger's expected payoff is

$$U_z(q) = (B_z + R_z)(q - q_z^*),$$

so appealing is privately optimal exactly when continuation is socially optimal, with common indifference at $q = q_z^$.*

Proof. Substitute (3) into $U_z(q) = qR_z - (1 - q)B_z - m_z$. The expression vanishes at $q = q_z^*$ and has strictly positive slope $B_z + R_z$. Hence its sign is the sign of $q - q_z^*$, which equals the sign of $G(z, q)$ by single crossing. $\square$

Corollary C.3 (The implemented bond-only threshold). *If successful appeal returns the principal plus $R = B/2$ and there is no additional private cost, then*

$$U(q) = B \left(\frac{3q}{2} - 1 \right), \quad q^{\text{private}} = \frac{2}{3}.$$

With private cost $m > 0$, the threshold becomes $2/3 + 2m/(3B)$ whenever that quantity is at most one. The flat implemented ratio therefore matches the planner threshold uniformly over $q \in [0, 1]$ only when the aligned social threshold is exactly $2/3$ (or the cost-adjusted value). On restricted support, different thresholds can induce the same observed actions simply because no history lies between them. Part 2's no-workload-free-optimum result rules out assuming this coincidence without calibration.

The theorem separates two levers that the current mechanism couples. The principal B_z can be chosen to cover triggered work or deter capital-poor spam; the reward R_z selects a private threshold. Equation (3) does not prove that the resulting transfer is budget-balanced, collusion-resistant, or legally observable. Those are additional mechanism constraints.

C.2 The price of threshold misspecification

The Bellman-gap identity gives a direct calibration loss rather than merely a classification error.

Proposition C.4 (One-cell contribution to threshold regret). *Suppose in one occupied cell z that q has density f_z and $G(z, q) = \kappa_z(q - q_z^*)$ for $\kappa_z > 0$. If private incentives induce the threshold policy $q > t_z$, this cell's conditional contribution to implementation regret is*

$$\kappa_z \int_{\min\{t_z, q_z^*\}}^{\max\{t_z, q_z^*\}} |q - q_z^*| f_z(q) dq.$$

For $q \sim \text{Uniform}[0, 1]$, this reduces to

$$\frac{\kappa_z}{2} (t_z - q_z^*)^2.$$

The full ladder regret is obtained by weighting such losses by the endogenous occupancy of each history, as in Lemma A.2.

Proof. The private and planner actions differ exactly between their two thresholds. At every such q , Corollary A.3 assigns Bellman gap $|G(z, q)| = \kappa_z |q - q_z^*|$. Integration gives the first formula; direct integration under the uniform density gives the second. $\square$

D Several Appellants: First-Valid-Transaction Preemption

An appeal is a discrete public good when several actors value continuation, but the contract's state transition is exclusive: only the first valid state-changing transaction can trigger a given successor. A simultaneous volunteer benchmark that ignores ordering has mixed-equilibrium appeal probability

$$p_N = 1 - (k/v)^{1/(N-1)}, \quad \mathbb{P}(\text{no appeal}) = (k/v)^{N/(N-1)} \rightarrow k/v$$

for effective cost $0 < k < v$ [5]. That calculation isolates free-riding, but it omits the preemption race created by transaction ordering. We now solve a reduced game that includes both forces.

D.1 A first-valid-transaction entry game

Definition D.1 (Preemptive appeal-entry game). There are $N \geq 2$ identical potential appellants. Each simultaneously chooses *submit* or *wait*. If at least one submits, one submission is selected uniformly as the first valid state-changing appeal and all agents receive continuation value $v \geq 0$. Every submitter incurs race cost $g \geq 0$, whether or not selected. The selected submitter additionally receives exclusive expected net prize $s \geq 0$, which can aggregate its success-contingent transfer and trigger-specific private value after expected bond exposure. If nobody submits, every payoff is zero.

The uniform ordering rule is a transparent reduced form, not a claim about a particular sequencer. Endogenous fee bids are treated separately below.

Theorem D.2 (Symmetric preemption equilibrium). Assume $v + s > 0$. The preemptive entry game has a unique symmetric equilibrium submission probability $p_N \in [0, 1]$:

$$p_N = \begin{cases} 0, & g \geq v + s, \\ 1, & g \leq s/N, \\ \text{the unique } p \in (0, 1) \text{ satisfying } v(1-p)^{N-1} + s \frac{1-(1-p)^N}{Np} = g, & s/N < g < v + s. \end{cases}$$

In the interior regime,

$$\mathbb{P}(\text{no effective appeal}) = (1 - p_N)^N > 0,$$

while the expected number of redundant submissions is

$$Np_N - [1 - (1 - p_N)^N] > 0.$$

Thus the same equilibrium can exhibit both underprovision and congestion. In the full-entry regime, exactly $N - 1$ submissions are redundant.

Proof. If each of the other $N - 1$ actors submits with probability p , let $X \sim \text{Binomial}(N - 1, p)$. Submitting rather than waiting changes expected payoff by

$$D_N(p) = v(1-p)^{N-1} - g + s \mathbb{E} \left[\frac{1}{X+1} \right] = v(1-p)^{N-1} - g + s \frac{1-(1-p)^N}{Np},$$

where the last ratio is continuously extended to one at $p = 0$. The first term is decreasing in p , and the expectation is decreasing because X is stochastically increasing while $x \mapsto 1/(x+1)$ is decreasing. Hence D_N is strictly decreasing. Its endpoint values are $D_N(0) = v + s - g$ and $D_N(1) = s/N - g$, proving the three symmetric-equilibrium regimes. The last two formulas follow because an effective appeal occurs once whenever at least one submission is made, whereas the expected number submitted is Np_N . $\square$

The theorem does not say that the interior equilibrium is socially optimal: v and s are private payoffs, while g can include both real resource cost and transfers. It says that “first transaction wins” changes the diagnosis. Permissionless entry can fail to produce an appeal and can simultaneously burn resources in losing submissions.

D.2 Priority competition as an all-pay limit

Transaction-ordering opportunities can induce priority-fee competition in blockchain systems [4]. To isolate its strongest rent dissipation effect, suppose N capital-eligible racers value one exclusive trigger rent at the same $s > 0$, submit priority expenditures $b_i \geq 0$, and forfeit those expenditures whether or not they obtain first position. This is the symmetric complete-information all-pay auction [1].

Proposition D.3 (Symmetric all-pay rent dissipation). The symmetric atomless equilibrium has cumulative distribution

$$F(b) = \left(\frac{b}{s} \right)^{1/(N-1)}, \quad 0 \leq b \leq s.$$

Every racer has zero expected net trigger rent, each spends s/N in expectation, and total expected priority expenditure equals the entire prize s .

Proof. A bid b wins with probability $F(b)^{N-1}$ and therefore earns $sF(b)^{N-1} - b$. Equating this payoff to its zero value at the lower endpoint gives the displayed distribution. Its mean expenditure is

$$\int_0^s [1 - F(b)] db = s - s \frac{N-1}{N} = \frac{s}{N}.$$

Summing over racers proves the claim. $\square$

Real ordering may be winner-pay, partially reverting, continuous-time, private-information, or sequencer-controlled; Proposition D.3 is therefore a stress-test limit, not a deployed gas estimate. It shows why making a trigger prize large can transform under-entry into an ordering rent rather than into costless reliable provision.

Corollary D.4 (Exclusive trigger slots remove this coordination failure). *If an unpredictable protocol lottery gives exactly one eligible actor an exclusive appeal slot, followed by fallback slots after observable timeouts, the within-slot choice reduces to the binary problem of Theorem C.2. Under that theorem's calibrated single-crossing assumptions, each available slot holder takes the planner's action. Availability, bribery, fallback delay, and selection manipulation remain separate design constraints.*

This corollary is a mechanism-design candidate, not a description of the current protocol. It illustrates why “anyone may appeal” and “someone will appeal whenever it is valuable” are not equivalent statements.

D.3 Capital-constrained participation

The bond is not only a payoff coefficient; it must be posted before the appeal resolves. Let K_i denote actor i 's liquid appeal capital. Suppose its non-bond participation cost is $m_i(B) = m_i^0 + \lambda_i B$ with $m_i^0, \lambda_i \geq 0$, its private transaction value is x_i , and the net success reward is $R = \zeta B$.

Proposition D.5 (Capital-adjusted threshold and availability). *A risk-neutral actor can appeal only if $K_i \geq B$, and conditional on that feasibility its strict participation condition is*

$$q_i > q_i^{\text{cap}}(B) = \frac{1 + \lambda_i + (m_i^0 - x_i)/B}{1 + \zeta}. \quad (4)$$

For the implemented $\zeta = 1/2$, zero private value, and zero participation cost, this reduces to $2/3$. It does not reduce to $2/3$ when capital has an opportunity cost or other private terms are present.

If N otherwise-willing actors have independent liquid-capital laws and $F_i^<(B) = \mathbb{P}(K_i < B)$, the probability that capital alone permits at least one appeal is

$$A_N(B) = 1 - \prod_{i=1}^N F_i^<(B). \quad (5)$$

Consequently $A_N(B)$ is nonincreasing in the bond and is an upper bound on the effective-appeal probability before beliefs, race costs, or ordering are applied.

Proof. Substitute $R = \zeta B$ and $m_i(B)$ into (1); the result is $(1 + \zeta)Bq_i - B(1 + \lambda_i) + x_i - m_i^0$, whose positive-sign condition is (4). An appeal is capital-feasible exactly when at least one $K_i \geq B$. Independence makes the complementary probability $\prod_i F_i^<(B)$, proving (5) and monotonicity. $\square$

For validator appeals, the pinned nominal jury sequence implies

appeal jury n'	7	13	25	49	97	193	385	769
B_{val}/W	7	13	25	49	97	193	385	769

on paths that reach those tiers. The last displayed bond locks about 109.9 times the capital of the first. The risk-neutral bond-only threshold can therefore remain numerically unchanged while the feasible appellant set shrinks sharply.

E Within-Panel Coordination and the Schelling Point

The stopping game decides whether another experiment is purchased. The round game decides how selected validators use the experiment they have already been paid to run.

E.1 What validator fees reward

Fix a round with committee size n . Let H be the event that the round is determined, d its outcome on H , $u = \mathbb{P}(H^c)$, and $q_i = \mathbb{P}(V_i = d \mid H)$ validator i 's conditional probability of matching that outcome. A determined-round majority validator earns $+W$, a minority validator pays κW , and every validator earns $+W$ when the round is undetermined.

Theorem E.1 (Outcome alignment is what gets paid). *The expected per-round fee profit is*

$$\mathbb{E}[\pi_i] = (1 - u)W[(1 + \kappa)q_i - \kappa] + uW.$$

With the implemented symmetric coefficient $\kappa = 1$, the determined-round component is $W(2q_i - 1)$. This is positive exactly when the validator is more likely than not to match the realized outcome; it is not a truth-contingent payment.

Proof. On a determined round, the validator receives W with probability q_i and $-\kappa W$ otherwise. On an undetermined round it receives W . Condition on the two cases. $\square$

For a separate stylized link to Part 1, let $p_i = \mathbb{P}(V_i = D_\infty \mid H)$ and $\varepsilon = \mathbb{P}(d \neq D_\infty \mid H)$. If the validator's signal and the rest-of-panel error event are conditionally independent given D_∞ and H , and the validator's pivotal effect is neglected, then

$$q_i = p_i(1 - \varepsilon) + (1 - p_i)\varepsilon.$$

Corollary E.2 (Conditional ambiguity discount). *Under that additional model and $\kappa = 1$,*

$$\mathbb{E}[\pi_i] = (1 - u)W(2p_i - 1)(1 - 2\varepsilon) + uW.$$

The relation is a conditional signal-model calculation, not a consequence of the fee rule or a semantic-accuracy guarantee.

E.2 The simultaneous round game

Definition E.3 (Round coordination game). An odd committee C plays a simultaneous binary-action game. Validator i observes its private execution verdict V_i and chooses $a_i \in \{0, 1\}$. It receives $+W$ when a_i matches the majority and $-W$ otherwise. Commit-reveal blocks observation of other committee votes before commitment; it does not block prior conventions, external communication, or collusion.

Proposition E.4 (A sufficient condition for informative voting). *Suppose every validator other than i votes its signal and, conditional on each value of V_i , their strict majority is at least as likely to equal V_i as to oppose it. Then voting V_i is a best response, strict when the inequality is strict for every signal type.*

Proof. Under the stated condition, matching the signal is at least as likely to place i in the majority as opposing it. The symmetric $+W/-W$ payoff makes that probability comparison the best-response comparison. $\square$

Proposition E.5 (Uninformative conformity equilibria). *The profiles in which every validator always plays 0 and in which every validator always plays 1, independently of its execution, are strict Nash equilibria of the one-shot round game.*

Proof. Against a unanimous profile, unilateral deviation changes a $+W$ majority payoff into a $-W$ minority payoff without changing the majority. $\square$

Part 1's product evaluator law does not select among these equilibria. It describes ballot draws when evaluators follow the declared execution rule; it does not turn following that rule into a dominant strategy.

E.3 Engineered focality

Coordination games require an equilibrium-selection story. In Schelling’s terminology [9], a focal point is salient for reasons not contained in the bare payoff matrix. OD supplies a canonical common reference: execute the same intelligible contract under the same equivalence principle and vote the resulting verdict. This is engineered focality in the lineage of SchellingCoin-style mechanisms [2], with auditable computation replacing unaided convention.

The claim must remain modest. Execution is the protocol-provided focal signal, not the only convention actors could share. Commit–reveal removes intra-round vote observation, randomized membership makes targeted coordination harder under additional cryptographic assumptions, and successful appeals can claw back a non-focal round. None of these properties eliminates off-chain coordination by theorem.

E.4 Corrective credibility as the cross-layer estimand

Definition E.6 (Corrective credibility). Let D_∞ be the Part 1 population-reference decision. Conditional on a population-inconsistent provisional round, define

$$\begin{aligned} a &= \mathbb{P}(\text{effective appeal} \mid d \neq D_\infty), \\ c &= \mathbb{P}(\text{corrective clawback} \mid \text{effective appeal}, d \neq D_\infty), \\ \chi &= \mathbb{P}(\text{effective appeal and corrective clawback} \mid d \neq D_\infty). \end{aligned}$$

The scalar χ is the round’s *corrective credibility*.

The factor a is generated by capital feasibility, appellant beliefs, and the preemption equilibrium. The factor c is generated by the successor panel, generator transition, and protocol success classification. Neither is supplied by Part 1’s one-panel theorem.

Proposition E.7 (Participation–correction decomposition). *Whenever the conditioning event has positive probability,*

$$\chi = ac, \quad 1 - \chi = (1 - a) + a(1 - c). \quad (6)$$

For an alternative mechanism with factors (a', c') , the exact change is

$$\chi' - \chi = c(a' - a) + a(c' - c) + (a' - a)(c' - c).$$

On a labeled calibration sample containing $E > 0$ inconsistent provisional rounds, $A > 0$ effective appeals, and C corrective clawbacks, the plug-in factorization satisfies

$$\widehat{a}\widehat{c} = (A/E)(C/A) = C/E = \widehat{\chi}.$$

Thus direct estimation identifies total credibility, while the factorization locates whether participation or conditional correction is the bottleneck.

Proof. The product identity is the conditional-probability chain rule. The failure identity partitions an inconsistent round into no appeal and an appeal that is not corrective. Expanding $a'c' - ac$ proves the comparison formula, and the sample identity follows by cancellation. $\square$

Consider the auxiliary game in which a population-inconsistent round is clawed back with probability χ , setting all its round payoffs to zero. Write $\mu(v) = \mathbb{P}(D_\infty = 1 \mid V_i = v)$ and let k be the number of the other $n - 1$ validators who choose 1.

Lemma E.8 (Exact action advantage). *The expected payoff advantage of action 1 over action 0 is*

$$\Delta(k) = \begin{cases} -2W[(1 - \mu) + \mu(1 - \chi)], & k \leq (n - 3)/2, \\ W\chi(2\mu - 1), & k = (n - 1)/2, \\ +2W[\mu + (1 - \mu)(1 - \chi)], & k \geq (n + 1)/2. \end{cases}$$

The game has strict strategic complementarities for every $\chi \in [0, 1]$.

Proof. Outside the pivotal case, the majority is fixed and the two actions exchange the positive and negative majority-alignment payoffs. A population-inconsistent majority survives with probability $1 - \chi$. At the pivotal count either action places i in the majority, so only the posterior probability that the chosen outcome equals D_∞ remains. Subtraction gives the three expressions. Both increments across the pivotal count equal $W(2 - \chi) > 0$. $\square$

Corollary E.9 (Laplacian implication of corrective credibility). *Under the stipulated Laplacian belief that k is uniform on $\{0, \dots, n - 1\}$,*

$$\bar{\Delta}(\mu) = \frac{1}{n} \sum_{k=0}^{n-1} \Delta(k) = W\chi(2\mu - 1).$$

For $\chi > 0$, the unique Laplacian best response away from posterior ties is the population-posterior mode. Its selection strength is

$$Wac |2\mu - 1|.$$

For $\chi = 0$, this selection criterion is silent.

Proof. There are $(n - 1)/2$ values of k on either non-pivotal side and one pivotal value. Pairing the two non-pivotal expressions in Lemma E.8 gives $2W\chi(2\mu - 1)$ per pair. Adding the pivotal term and dividing by n gives $W\chi(2\mu - 1)$. Its sign proves the best-response statement; substitute $\chi = ac$ for the final expression. $\square$

This calculation is related to risk-dominance and global-games reasoning [6, 3, 8], but it is not a global-games uniqueness theorem. For every $\chi < 1$, action 1 is profitable against a sufficiently large 1 majority and action 0 is profitable against a sufficiently large 0 majority, whatever μ . The required dominance regions are absent, and the conformity equilibria survive. The result identifies a selection channel: rational corrective appeals make the execution-based focal convention more attractive under a stated belief, but the focal convention does not itself guarantee that appeals occur.

F The GenLayer Mechanism as an Instance

F.1 Protocol refinement map

The canonical OD state graphs in `opt-dem-spec` (pinned through the `od-tla-spec` submodule) verify the control-flow skeleton against a pinned Solidity baseline. The fee-distribution simulator is the executable specification of transaction-level accounting. They validate different objects: the graph does not model validator-set composition or fees, and the fee simulator does not supply statistical transition probabilities. The repository accompanying this draft pins `od-tla-spec` at `d113abb`, its nested `opt-dem-spec` at `d06cb125`, and the top-level fee simulator at `93e965c`; every protocol claim is revision-scoped to those artifacts.

Paper object	OD realization	Abstraction
H_r	public status, tally, round, committees, appeal window	reduced models may omit rotations, idleness, and category detail
d_r	provisional Accepted/non-Accept projection	the protocol has several non-Accept categories
Continue	an effective validator appeal or an automatic leader/timeout successor	not every successor is purchased by the same appellant game
Stop	no valid appeal before expiry, then readiness/finalization	READY_TO_FINALIZE is virtual and finalization requires a call
S_h	protocol-defined appeal success by outcome-category comparison	not equivalent to agreement with D_∞
preemption and capital	first valid state-changing appeal wins; every sender must fund its transaction and the winner posts the bond	uniform ordering, all-pay priority, and independent capital are declared stress models, not sequencer facts
a_{r+1}	evaluator work, delay, burns, and other real incremental cost	transfers are not social resource cost

This is a refinement map, not an implementation-equivalence claim. In particular, appeals from **Revealing**, automatic leader appeals, leader-timeout successors, failed-appeal restoration, and voided/hanging paths must retain their full protocol state when they change the next kernel or payoff.

F.2 The executable fee mechanism

Two fee units parameterize the covered transaction-level accounting: L prices the leader time budget and W one validator time budget. The appeal bond depends on the work triggered:

$$B_{\text{val}} = n'W, \quad B_{\text{lead}} = (\rho + 1)(L + mW), \quad B_{\text{lt}} = (\rho + 1)(L + n_{\text{cur}}W).$$

A validator appeal re-evaluates the existing proposal with n' leaderless appeal validators. A leader appeal triggers a new normal round of size m and budgets up to ρ rotations. A leader-timeout appeal rotates the current committee rather than moving to the next statistical size tier.

The pinned fee simulator models 13 round labels and checks 24 invariants over generated paths. The accounting results used here reduce to three economic properties:

Mechanism Invariant F.1 (Cost of contention). A failed appeal costs its appellant at least the posted bond on every covered path.

Mechanism Invariant F.2 (No profit from minority grieving). A coalition assigned the minority payoff has non-positive net earnings after the covered costs, burns, and slashes.

Mechanism Invariant F.3 (Bounded grieving amplification). On a covered failed-appeal path, financial damage imposed on other participants does not exceed the appellant's own cost.

Proposition F.4 (GenLayer bonds fund work, not Bellman optimality). *Each formula above covers the worst-case payroll of the work its branch triggers under the executable fee specification. This establishes self-funding and the covered-path accounting invariants. It does not imply that the induced appeal policy minimizes J_Λ .*

Proof. The three formulas are the corresponding validator payroll, normal-round payroll with rotations, and current-committee timeout payroll with rotations. Pathwise conservation in the simulator closes their accounting. The final claim follows both from Proposition B.1 and directly from the fact that none of these formulas contains the workload loss, posterior value of information, or private external transaction value. $\square$

On successful appeal the deployed rule returns $1.5B$, hence a net reward $R = B/2$ and the bond-only threshold $q > 2/3$ from Corollary C.3. Because B cancels, increasing panel size and the corresponding bond does not change this threshold for an unconstrained risk-neutral actor with zero capital cost. Proposition D.5 shows the missing qualification: the nominal validator-appeal bond grows from $7W$ to $769W$ along the documented jury sequence, so feasibility and opportunity cost can change by tier even when the displayed reward ratio does not. The artifacts contain no appellant capital distribution, borrowing market, or live ordering-cost model. The economic stopping frontier of Part 2 independently changes with evaluator cost, latency, workload loss, future panel law, and remaining options.

F.3 What the current safeguards do and do not buy

Retroactive **SKIP** accounting removes the round reward from an overturned herd on covered paths. Commit-reveal prevents a validator from waiting to copy revealed committee votes. The finite, capacity-constrained round array bounds the number of escalation opportunities on any realized path. These are meaningful constraints on manipulation, but they do not prove any of the following:

- that a socially valuable appeal receives a willing appellant;
- that a privately profitable appeal has positive social value;
- that a protocol-successful appeal is population-corrective;
- that the resulting a or c in Definition E.6 is bounded away from zero;
- that a fixed Byzantine network share cannot influence several correlated or identity-reusing committees.

The leader payoff supplies a narrower alignment result. A clear normal round pays the leader $L + W$, an undetermined round pays L , and a timeout pays zero before branch-specific compensation. Holding external transaction value fixed, leaders prefer clarity in fee terms. Whether the one-unit difference W prices the external value of delay is an empirical question.

F.4 Revision-scoped caveats

The fee simulator’s statistical schedule tops out at 1000, whereas the archived FeeManager revision contains 18 nominal size entries through 1537. The contract has no stand-alone “eight appeals” counter: feasibility depends on the current round and the available unconsumed validator count. In the documented $N = 1543$ scenario, eight appeal juries through size 769 can lead to a 1535-member normal successor, whose strict-majority threshold is 768 under the archived voting contract, not 772. The companion executable specification additionally applies a minus-two adjustment after consecutive unsuccessful appeals. Validator identities can be reused in normal successors, the leader occupies a voting seat, idle seats can be replaced, and ballots reset. The voided appeal path remains economically undefined in the pinned artifacts: the challenged state may be restored while reward classification and bond routing require a protocol decision. All accounting claims in this section are therefore restricted to the non-voided paths checked by the executable specification.

G How to Calibrate and Test Decentralized Optimal Stopping

The positive theorem is operational only after its primitives are estimated. A defensible evaluation should proceed in the following order.

1. **Declare the social problem.** Fix a workload law Λ , the population-reference target, loss $L(P)$, evaluator work prices, latency price, admissible policies, and any hard risk budget. Without these declarations there is no unique “optimal” stopping rule.
2. **Estimate the transition kernels.** For every modeled history cell, estimate the distribution of successor categories under the actual committee-selection and execution policy. Part 1 certificates can support declared future-panel statements; one observed live tally is not by itself such a certificate.
3. **Solve the planner first.** Run the finite-horizon backward recursion from Part 2 to obtain $G_r(h)$ and π^* . Test whether a proposed scalar success statistic has the single-crossing property rather than assuming it.
4. **Estimate private behavior separately.** Measure calibrated success beliefs, monitoring and capital cost, liquid-capital availability, private transaction value, submission/ordering costs, and the number and information structure of potential appellants. Estimate the *joint* law of social loss and private value needed by Proposition B.3; their marginals do not determine implementation regret. Then solve or bound the induced equilibrium policy π^{eq} .
5. **Report implementation, not appeal volume.** Use Lemma A.2 to report premature-stop mass, excessive-escalation mass, Bellman-gap-weighted regret, statistical-risk feasibility, race waste, capital exclusion, and corrective credibility $\chi = ac$. On labeled errors, report both C/E and its decomposition into A/E and C/A .
6. **Recalibrate out of sample.** A state-dependent transfer schedule must be committed before evaluating the histories used to test it. Adaptive changes to workload cells, margins, or model families require new multiplicity and strategic-manipulation analysis.

Observed appeals are selected data: success is observed only after somebody finds appeal privately worthwhile. Naively estimating $q(h)$ from historical appeals therefore confounds the transition law with the old incentive rule. Randomized shadow quotes, bounded experimental variation in rewards, or an explicit selection model are possible identification strategies. Any live experiment must additionally bound the value placed at risk.

The Part 1 real-LLM pilot does not fill this gap. It demonstrates an end-to-end evaluator matrix and measures within-task coordination for a small diagnostic sample. It contains neither sequential successor panels under the deployed sampler nor appellant utilities, so it cannot estimate π^{eq} , q_z^* , or implementation regret.

Supplementary Material

S1 Deterministic Numerical Illustrations

The two main-text figures visualize exact expressions from the appendices. They are diagnostic slices of the theory, not estimates from deployed appeals, validator behavior, or transaction losses. No randomness enters their construction.

S1.1 Implementation and preemption

The left panel of Figure 1 evaluates the planner threshold q^* against a private threshold fixed at $2/3$, the bond-only value induced by a successful gross return of $1.5B$. For every pair (q, q^*) on the plotted grid, the planner continues when $q > q^*$ and the private trigger appeals when $q > 2/3$. The four regions record agreement or the direction of the action mismatch.

The right panel uses the symmetric preemption equilibrium of Theorem D.2 with

$$N = 10, \quad \nu = 1, \quad s = 0.3,$$

and submission cost g varying from zero to 1.4 . For interior points, the unique root is obtained by bisection from the strictly decreasing equilibrium condition. The plot reports the equilibrium submission probability, the probability of no effective appeal, and the expected number of redundant submissions.

S1.2 Bond scale and capital availability

The left panel of Figure 2 uses the nominal validator-appeal jury sizes

$$7, 13, 25, 49, 97, 193, 385, 769$$

and the bond identity $B_{\text{val}} = n'W$. The right panel is a sensitivity calculation under independent lognormal liquid capital with median $100W$ and log-scale standard deviation one. It evaluates

$$A_N(B) = 1 - [\mathbb{P}(K < B)]^N$$

for $N \in \{1, 5, 20\}$. Independence and the chosen distribution are illustrative assumptions. The panel should not be read as a capital audit or a prediction of appeal incidence.

Changing these parameters affects the displayed magnitudes but not the proved comparative statics: capital-only availability is nonincreasing in the bond, and the preemption game's interior contains both non-provision and duplicate entry.

S2 Reproducibility

The repository contains the deterministic figure generator and unit tests in the Part 4 simulations directory. From the repository root, the numerical artifacts can be regenerated with

```
python3 papers/part4-gametheory/simulations/mechanism_design_figures.py
python3 -m unittest discover \
  -s papers/part4-gametheory/simulations -p 'test_*.py'
```

The generator requires NumPy and Matplotlib. It writes the two PNG figures used by the manuscript. The test suite checks the Bellman-gap identity on finite examples, the calibrated reward signs, the two-thirds threshold, the preemption equilibrium regimes, all-pay rent dissipation, capital availability, and the joint-dispersion expression. All calculations are local and deterministic; no API key, private dataset, or network service is required.